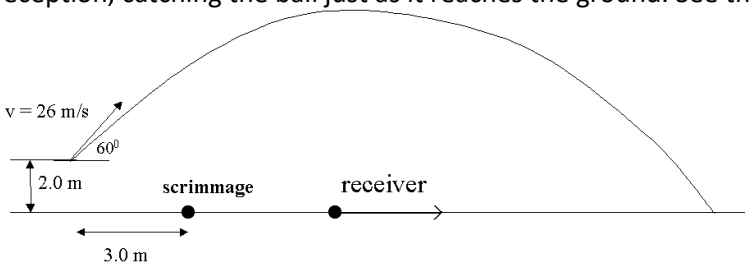
Problem Solving Review Questions

1. A football quarterback attempts a pass to one of the receivers. As the ball is snapped, the receiver leaves the line of scrimmage and runs directly down field. The quarterback releases the ball 2.0 s later and from a position 3.0 m behind the line of scrimmage. He throws the ball with a speed of 26 m/s at an elevation of 60^o above the horizontal. The receiver makes a diving reception, catching the ball just as it reaches the ground. See the diagram below.



   a. What is the time of flight of the football? [4.7 s]
   b. What is the average speed of the receiver? [8.7 m/s]

2. An arrow is shot vertically upward with an initial speed of 25 m/s. When it's exactly halfway to the top of its flight, a second arrow is launched vertically upward from the same spot. The second arrow reaches the first arrow just as the first arrow reaches its highest point.
   a. What is the launch speed of the second arrow? [26.5 m/s]
   b. What maximum height does the second arrow reach? [35.8 m]

3. A truck travels at a constant speed of 28.0 m/s in the fast lane of a two-lane highway. It approaches a stationary car stopped at the side of the road. When the truck is still 1.2×10^2 m behind the car, the car pulls out into the slow lane with an acceleration of 2.6 m/s^2.
   a. How long will it take the truck to pass the car? [5.9 s]
   b. How far will the car have travelled when the truck passes it? ~~[1.6 x 10^2 m]~~ [45 m]
   c. If the car were to maintain this acceleration, what would be its speed when it overtakes the truck? [41 m/s]

4. A clock has a second hand 15 cm long, from centre to tip. (Hint: the velocity of the tip of the second hand is always tangent to the circle)
   a. Draw a labelled diagram to represent this situation
   b. Calculate the instantaneous speed of the tip of the second hand. [1.57 cm/s]
   c. What is the instantaneous velocity of the tip at 15 s? [1.57 cm/s [down]]
   d. What is its change in velocity between 30 s and 45 s? [2.2 cm/s [up 45^o right]]
   e. What is its average acceleration during the same time interval? (1 mark) [0.15 cm/s^2 [up 45^o right]]

Textbook Questions:

Problems: page 25 #3, p 29 #3, 6, 10, 11, p 35 #10, 11, 12, p 58 #71, 72, 74, p 43 #1-3, 9, p 49 #1, 3, 5, 7,

Multiple Choice, True/False & Short Answer: p 53 #6, 7, 10-15, p 54 #2, 5, 6, 11, 13, 14, p 58#19, 20, 22, 27, 31, 32, 33, 34

work on following pages

Concepts in Kinematics

1. How many people are there in this room? N How many significant digits is this? ∞

   N = count; exact

2. Explain the difference between the following symbols:

   a. v_{av}, Δv, v_i

   $v_{av} = \frac{\Delta d}{\Delta t}$: average velocity

   $\Delta v = v_f - v_i$: change in velocity

   v_i : initial velocity

   b. $\vec{d}$, $\Delta\vec{d}$

   d : position from an origin

   Δd : displacement (change in position)

3. Two toddlers are 10 m apart and are moving towards each other, one travelling at 0.4 m/s and the other at 0.25 m/s. The faster child started 2.2 s after the slower one. Which of the following statements are true at the time the two children meet? (s = slow; f = fast)

   a. $\Delta t_f = \Delta t_s + 2.2s$

   b. $\Delta t_s = \Delta t_f + 2.2s$.

   Explain your answer: Slow child starts first, so walks 2.2 s longer.

4. The following vectors show the velocity of a roller coaster at two different points in time. What is the direction of the acceleration? Explain.

   $\vec{v}_1$ $\vec{v}_2$

   a is down and right, along $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$.

5. The velocity of a dragonfly at two different times is shown by the vectors below. Draw a vector diagram that would allow you to calculate the dragonfly's change in velocity.

   $\vec{v}_f$ $\vec{v}_i$

   $\Delta\vec{v} = \vec{v}_f + (-\vec{v}_i)$

6. The position of a student at two points in time is shown by the vectors below. Draw a tip-to-tail diagram that would show $\vec{d}_f - \vec{d}_i$. What does this vector represent physically?

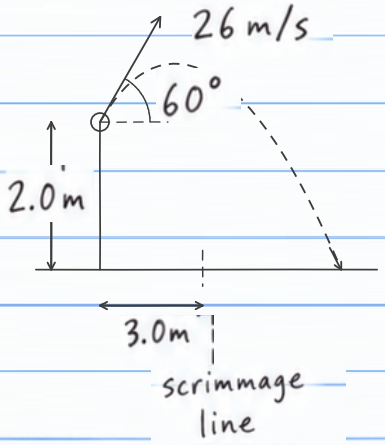
   $\vec{d}_f$ $\vec{d}_i$ O

   $\Delta\vec{d} = \vec{d}_f + (-\vec{d}_i)$

   displacement, from initial to final position

Review Questions

1) +y is up, +x is downfield



a) $v_{ix} = 26 \cos 60° = 13.0 \text{ m/s}$

$v_{iy} = 26 \sin 60° = 22.52 \text{ m/s}$

$\Delta d_y = -2.0 \text{ m}, \quad a_y = -9.81 \text{ m/s}^2$

$$\Delta d_y = v_{iy} \Delta t + \frac{1}{2} a_y \Delta t^2$$

$$-2.0 = 22.52 \Delta t - 4.905 \Delta t^2$$

$$4.905 \Delta t^2 - 22.52 \Delta t - 2.0 = 0$$

use quadratic formula

$\Delta t = -0.087 \text{ s}$ or 4.678 s

negative time is before the throw

$$\Delta t_{ball} = 4.7 \text{ s}$$

b) $\Delta d_{ball,x} = 13.0(4.6777) = 60.81 \text{ m}$

receiver starts 3.0 m ahead of the quarterback

$\Delta d_{receiver} = 60.81 - 3.0 = 57.81 \text{ m}$

$\Delta t_{receiver} = 4.6777 + 2.0 = 6.6777 \text{ s}$

$$v_{av} = \frac{57.81}{6.6777}$$

$$v_{av} = 8.7 \text{ m/s}$$

2) + is up

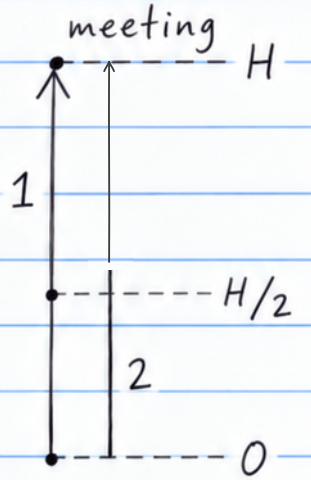
$\vec{v}_i = 25\ m/s$

at the top, $\vec{v}_f = 0$

$0 = 25^2 - 2(9.81)H$

$H = 31.86\ m$

$\Delta t_{top} = \frac{25}{9.81} = 2.548\ s$



halfway up:

$\Delta d = \frac{H}{2} = 15.93\ m$

$v_{half}^2 = 25^2 - 2(9.81)(15.93)$

$v_{half} = 17.68\ m/s$

$\Delta t_{half} = \frac{17.68 - 25}{-9.81} = 0.746\ s$

$2.548 - 0.746 = 1.802\ s$

a) $H = v_{i2}\Delta t - \frac{1}{2} g \Delta t^2$

$31.86 = v_{i2}(1.802) - 4.905(1.802)^2$

$v_{i2} = 26.5\ m/s\ [up]$

b) $0 = v_{i2}^2 - 2gH_2$

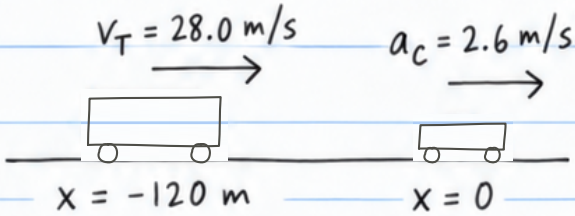
$H_2 = \frac{26.5165^2}{2(9.81)}$

$H_2 = 35.8\ m$

3) + is forward, car starts at $x = 0$

truck: $x_T = -120 + 28.0\,\Delta t$

car: $x_C = \frac{1}{2}(2.6)\,\Delta t^2$



same position when they pass

$1.3\,\Delta t^2 = -120 + 28.0\,\Delta t$

$1.3\,\Delta t^2 - 28.0\,\Delta t + 120 = 0$

use quadratic formula

$$\Delta t = \frac{28.0 \pm \sqrt{28.0^2 - 4(1.3)(120)}}{2.6}$$

$\Delta t = 5.904$ s or 15.634 s

a) truck passes first

$\Delta t = 5.9$ s

b) $\Delta d_C = \frac{1}{2}(2.6)(5.904)^2$

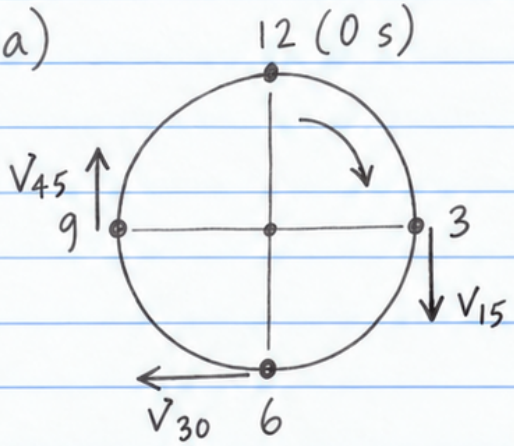
$\Delta d_C = 45$ m

c) use the later time when the car overtakes

$v_C = 0 + 2.6(15.634)$

$v_C = 41$ m/s

4a)



$r = 15$ cm

$T = 60$ s

b) $v = \dfrac{2\pi r}{T} = \dfrac{2\pi(15)}{60}$

$v = 1.57$ cm/s

c) tangent at 15 s points down

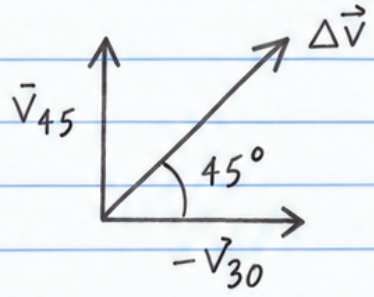
$V_{15} = 1.57$ cm/s [down]

d) $\Delta V = V_{45} + (-V_{30})$

V_{30} is left, so $-V_{30}$ is right

$|\Delta V| = \sqrt{1.57^2 + 1.57^2} = 2.22$ cm/s

$\theta = \tan^{-1}\left(\dfrac{1.57}{1.57}\right) = 45°$



$\Delta V = 2.2$ cm/s [up 45° right]

e) $a_{av} = \dfrac{\Delta V}{\Delta t} = \dfrac{2.22}{15}$

$a_{av} = 0.15$ cm/s^2 [up 45° right]

Kinematics Problems

1. A ball is thrown vertically upward from a 2nd story window with an initial velocity of 10 m/s. If a person in a third story window that is 5.0 m above the first window catches the ball on its way down, how long was the ball in the air? [1.2 s]

2. If the sum of two vectors equals zero, the magnitude of their difference is
   a) equal to the magnitude of either one
   b) equal to twice the magnitude of either one
   c) less than twice the magnitude of either one.

3. A turtle is moving with constant acceleration along a straight ditch. He starts his stopwatch as he passes a fence post and notes that it takes him 10s to reach a pine tree 10m farther along the ditch. As he passes the pine tree his speed is 1.2 m/s. How far was he from the fence post when he started from rest? (SIN 76) [ans: 8.0 m]

4. A British Concorde (BC) and a French Concorde (FC) flew in opposite directions around the earth (40 000 km). The BC covered half of its flight distance at a supersonic speed of 2500 km/h and the other half at a subsonic speed of 1000 km/h. The FC spent half of its flight time at 2500 km/h and the other half at 1000 km/h. Which Concorde completed the trip first, and by how many hours did it beat the other? [FC by 5.1 h]

5. The Easter bunny runs along a straight and narrow path with a constant speed of 25m/s. He passes a sleeping tortoise, which immediately starts to chase the bunny with a constant acceleration of 3.0×10^{-3} m/s^2. How long does it take the tortoise to catch the bunny? [4.6 h]

6. A person travelling between London and Windsor (200 km apart) wants to complete a round trip at an average speed of 100 km/h. On arriving at Windsor at the end of the first leg, she discovers that she has only averaged 75km/h. What must her average speed by on the return trip? (CAP Contest 1984) [150 km/h]

7. A car going 80.0 km/h overtakes a 1.20 km long train that is travelling at 65 km/h in the same direction on a track parallel to the road. How long does it take the car to pass the train? How long would it take if the vehicles were approaching each other travelling in opposite directions? [4.8 min; 0.50 min]

work on following pages

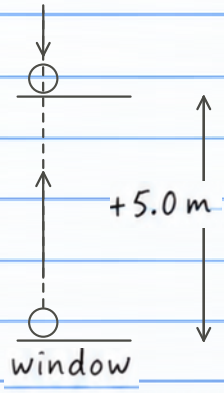
Problem Solving

1) + is up

$v_i = 10\ m/s$

$a = -9.81\ m/s^2$

$\Delta d = 5.0\ m$



$$\Delta d = v_i \Delta t + \frac{1}{2} a \Delta t^2$$

$$5.0 = 10\Delta t - 4.905\Delta t^2$$

$$4.905\Delta t^2 - 10\Delta t + 5.0 = 0$$

use quadratic formula

$$\Delta t = \frac{10 \pm \sqrt{100 - 4(4.905)(5.0)}}{9.81}$$

$\Delta t = 0.88\ s$ or $1.16\ s$

use the later time since it is caught going down

$\Delta t = 1.2\ s$

2) $\vec{A} + \vec{B} = 0$, so $\vec{B} = -\vec{A}$

$\vec{A}$ $\vec{B}$

(equal length)

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B}) = 2\vec{A}$$

$\vec{A}$ $-\vec{B}$

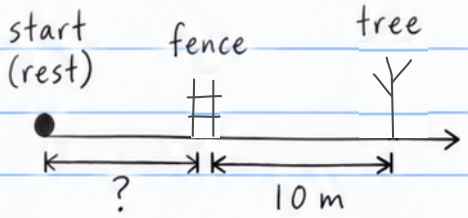
$2\vec{A}$

(b) twice the magnitude

3)

$$\Delta d = \frac{1}{2}(v_{fence} + v_{tree})\,\Delta t$$

$$10 = \frac{1}{2}(v_{fence} + 1.2)(10)$$



$$v_{fence} = 0.80 \text{ m/s}$$

$$a = \frac{1.2 - 0.80}{10} = 0.040 \text{ m/s}^2$$

start to fence:

$$v_{fence}^2 = v_{start}^2 + 2a\,\Delta d$$

$$\Delta d = \frac{0.80^2 - 0^2}{2(0.040)}$$

$\Delta d = 8.0$ m before the fence

4)

British: half the distance at each speed

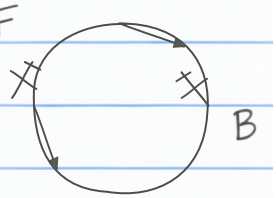
$$\Delta t_B = \frac{20000}{2500} + \frac{20000}{1000} = 28 \text{ h}$$

French: half the time at each speed

$$40000 = 2500\left(\frac{\Delta t_F}{2}\right) + 1000\left(\frac{\Delta t_F}{2}\right)$$

$$\Delta t_F = \frac{40000}{1750} = 22.86 \text{ h}$$

$$28 - 22.86 = 5.14 \text{ h}$$

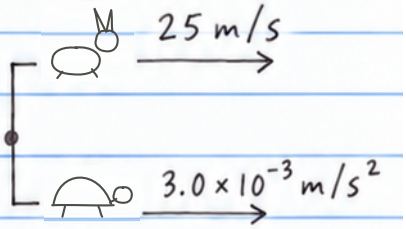


French Concorde, by 5.1 h

5) same position when they meet

$\Delta d_{bunny} = 25\Delta t$

$\Delta d_{tortoise} = \frac{1}{2}(3.0\times10^{-3})\Delta t^2$



$25\Delta t = 0.0015\Delta t^2$

$\Delta t(0.0015\Delta t - 25) = 0$

$\Delta t = 0 \quad \text{or} \quad 1.67\times10^4 \text{ s}$

0 is when they start, so use the later time

$\Delta t = 4.6 \text{ h}$

6) find the total time first

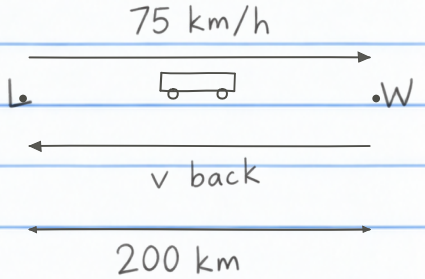
$\Delta t_{total} = \frac{(2\times200)}{100} = 4.0 \text{ h}$

$\Delta t_{out} = \frac{200}{75} = 2.67 \text{ h}$

$\Delta t_{back} = 4.0 - 2.67 = 1.33 \text{ h}$

$V_{back} = \frac{200}{1.33}$

$V_{back} = 150 \text{ km/h}$



7) + is the car's direction

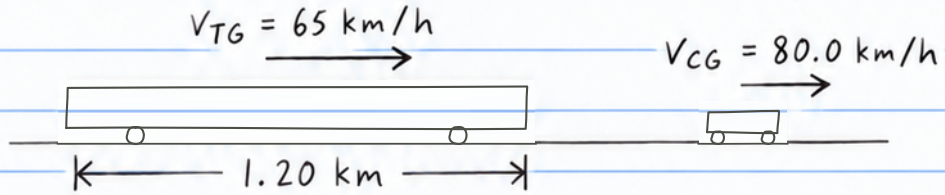
train length = 1.20 km

same direction:

$V_{CT} = V_{CG} - V_{TG} = 80.0 - 65 = 15 \text{ km/h}$

$\Delta t = \frac{L}{V_{CT}} = \frac{1.20}{15} = 0.080 \text{ h}$

$\Delta t = 4.8 \text{ min}$

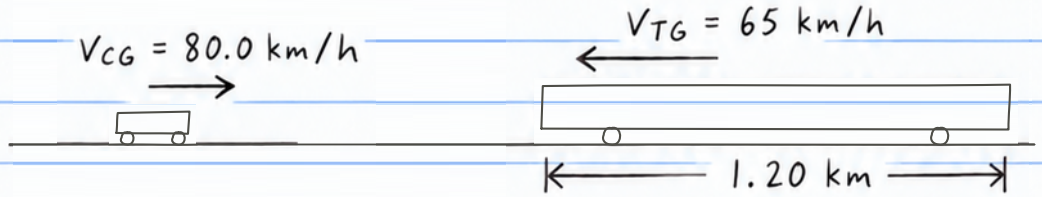


opposite directions:

$V_{CT} = 80.0 - (-65) = 145 \text{ km/h}$

$\Delta t = \frac{1.20}{145} = 0.00828 \text{ h}$

$\Delta t = 0.50 \text{ min}$ (about 30 s)



Mr. Harman
Physics

Relative Motion Worksheet

1. A passenger at the rear of a train traveling at 15 m/s relative to Earth throws a baseball with a speed of 15 m/s in the direction opposite the motion of the train. What is the velocity of the baseball relative to Earth as it leaves the thrower's hand?

2. A spy runs from the front to the back of an aircraft carrier at a velocity of 3.5 m/s. If the aircraft carrier is moving forward at 18.0 m/s, how fast does the spy appear to be running when viewed by an observer on a nearby stationary submarine?

3. A ferry is crossing a river. If the ferry is headed due north with a speed of 2.5 m/s relative to the water and the river's velocity is 3.0 m/s to the east, what will the boat's velocity relative to Earth be? (Hint: Remember to include the direction in describing the velocity.)

4. A girl at an airport rolls a ball north on a moving walkway that moves east. If the ball's speed with respect to the walkway is 0.15 m/s and the walkway moves at a speed of 1.50 m/s, what is the velocity of the ball relative to the ground?

5. Describe the motion of the following objects if they are observed from the stated frames of reference:
   a. a person standing on a platform viewed from a train traveling north
   b. a train traveling north viewed by a person standing on a platform
   c. a ball dropped by a boy walking at a speed of 1 m/s viewed by the boy
   d. a ball dropped by a boy walking 1 m/s as seen by a nearby viewer who is stationary

6. If the Green Goblin is riding his Goblin Glider in an effort to evade Spiderman. Spiderman is running at 10.0 m/s toward the front on top of a train traveling at 15.0 m/s East. Green Goblin decides to run at 5.0 m/s W across his glider that is moving at 23.0 m/s North in an attempt to jump off and avoid Spiderman's web. What is Green Goblin's speed relative to Spiderman so Spiderman can accurately shot his web?

7. A plane traveling north at 100.0 km/h through the air gets caught in a 40.0 km/h crosswind blowing west. This turbulence caused a beverage cart to brake free and begin rolling at 20.0 km/h toward the tail of the plane. What is the velocity of the cart relative to the ground? (you do not have to convert these since they are all the same unit)

Honors Only

8. A pet-store supply truck moves at 25.0 m/s north along a highway. Inside, a dog moves at 1.75 m/s at an angle of N 35.0° E. What is the velocity of the dog relative to the road?

9. Captain America is chasing Red Skull. He plans to throw his shield to knock down Red Skull but needs to know how fast Red Skull is moving relative to himself. If Red Skull is running at 3.50 m/s at an angle of 45.0° toward the left front of a truck that is moving at 12.5 m/s N, and Captain America is running at 4.00 m/s at an angle of 45.0° to the front left of a second truck moving South at 15.0 m/s, find the relative velocity for Captain America to hit Red Skull.

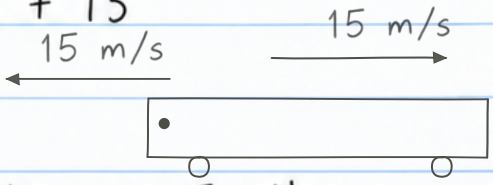
work on following pages

Relative Motion

1) + is the train's direction

$$V_{BG} = V_{BT} + V_{TG} = -15 + 15$$

$$\boxed{V_{BG} = 0 \text{ m/s}}$$

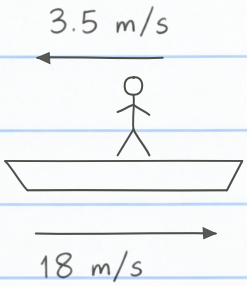


momentarily at rest relative to Earth

2) + is forward

$$V_{SG} = V_{SC} + V_{CG} = -3.5 + 18.0$$

$$\boxed{V_{SG} = 14.5 \text{ m/s [forward]}}$$



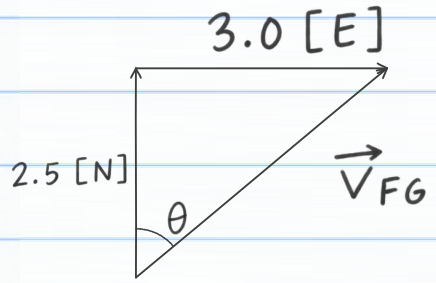
3) $V_{FG} = V_{FW} + V_{WG}$

$= 2.5\,[N] + 3.0\,[E]$

$|\vec{V}_{FG}| = \sqrt{2.5^2 + 3.0^2} = 3.91 \text{ m/s}$

$\theta = \tan^{-1}\left(\frac{3.0}{2.5}\right) = 50.2°$

$$\boxed{V_{FG} = 3.9 \text{ m/s [N 50° E]}}$$



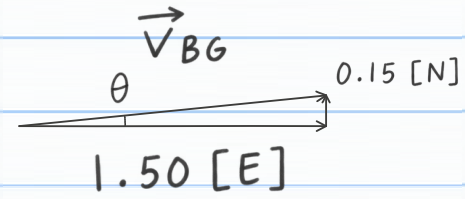
4) $V_{BG} = V_{BW} + V_{WG}$

$= 0.15\,[N] + 1.50\,[E]$

$|\vec{V}_{BG}| = \sqrt{0.15^2 + 1.50^2} = 1.51 \text{ m/s}$

$\theta = \tan^{-1}\left(\frac{0.15}{1.50}\right) = 5.71°$

$$\boxed{V_{BG} = 1.5 \text{ m/s [E 5.7° N]}}$$



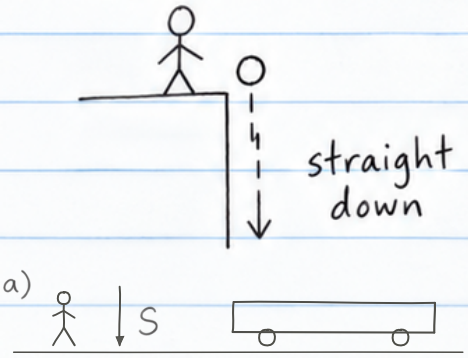
5) a) person appears to move south at the train's speed

b) train moves north

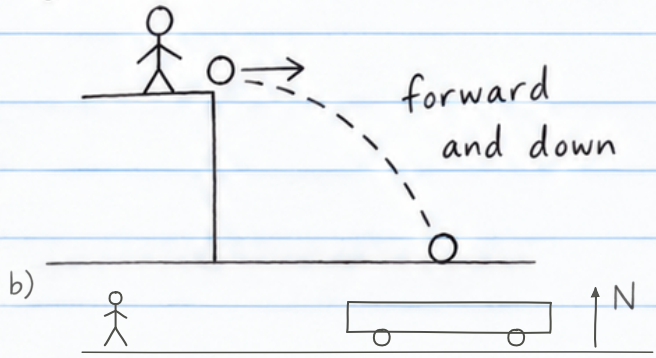
c) ball falls straight down relative to the boy

d) ball keeps moving forward at 1 m/s as it falls, so the stationary viewer sees a curved path

boy's view:



ground view:



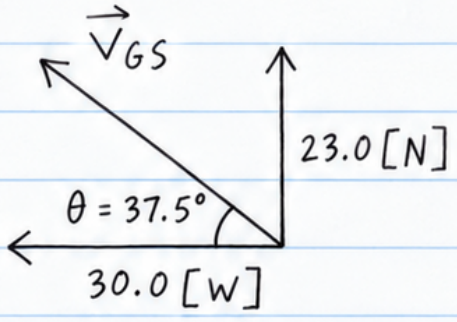
6) find each velocity relative to the ground first

$$\vec{V}_{SE} = 10.0\,[E] + 15.0\,[E] = 25.0\,[E]$$

$$\vec{V}_{GE} = 5.0\,[W] + 23.0\,[N]$$

$$\vec{V}_{GS} = \vec{V}_{GE} - \vec{V}_{SE}$$

$$= 30.0\,[W] + 23.0\,[N]$$



$$|\vec{V}_{GS}| = \sqrt{30.0^2 + 23.0^2} = 37.8 \text{ m/s}$$

$$\theta = \tan^{-1}\left(\frac{23.0}{30.0}\right) = 37.5^\circ$$

speed = 38 m/s

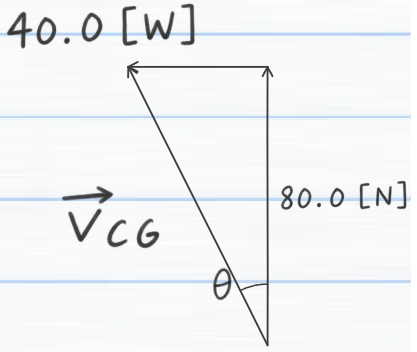
velocity = 38 m/s [W37°N] relative to Spiderman

S = Spiderman, G = Goblin, E = Earth

7) $\vec{V}_{CG} = \vec{V}_{CP} + \vec{V}_{PA} + \vec{V}_{AG}$

$= 20.0\,[S] + 100.0\,[N] + 40.0\,[W]$

$= 80.0\,[N] + 40.0\,[W]$



$|\vec{V}_{CG}| = \sqrt{80.0^2 + 40.0^2} = 89.4$ km/h

$\theta = \tan^{-1}\left(\frac{40.0}{80.0}\right) = 26.6°$

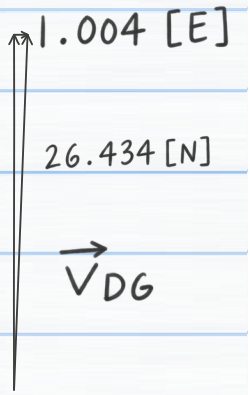
$\vec{V}_{CG} = 89.4$ km/h [N26.6° W]

C = cart, P = plane, A = air, G = ground

8) split the dog's velocity into components

$\vec{V}_{DT,N} = 1.75 \cos 35.0° = 1.434$ m/s

$\vec{V}_{DT,E} = 1.75 \sin 35.0° = 1.004$ m/s



$\vec{V}_{DG} = \vec{V}_{DT} + \vec{V}_{TG}$

$= (25.0 + 1.434)\,[N] + 1.004\,[E]$

$|\vec{V}_{DG}| = \sqrt{26.434^2 + 1.004^2} = 26.5$ m/s

$\theta = \tan^{-1}\left(\frac{1.004}{26.434}\right) = 2.17°$

$\vec{V}_{DG} = 26.5$ m/s [N2.17° E]

D = dog, T = truck, G = ground

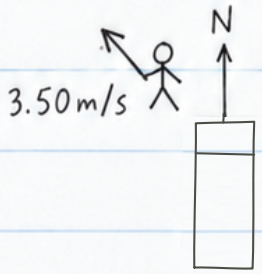
9) +x is east, +y is north

front-left depends on which way the truck is facing

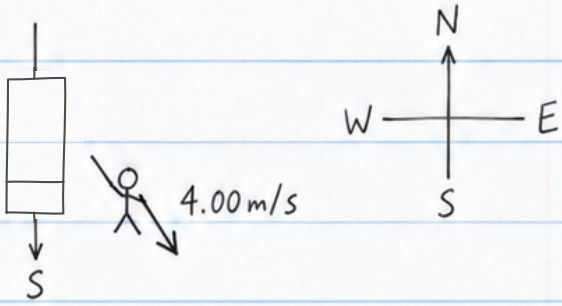
Red Skull's truck faces north, so he runs NW

Captain's truck faces south, so he runs SE

Red Skull:



Captain America:



Red Skull:

$V_{RG,x} = -3.50 \sin 45° = -2.475 \text{ m/s}$

$V_{RG,y} = 12.5 + 3.50 \cos 45° = 14.975 \text{ m/s}$

Captain America:

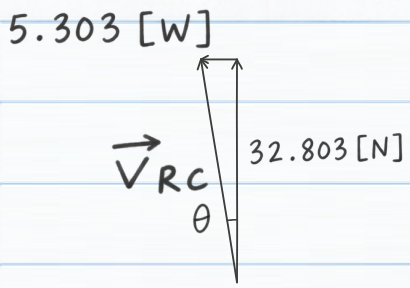
$V_{CG,x} = +4.00 \sin 45° = +2.828 \text{ m/s}$

$V_{CG,y} = -15.0 - 4.00 \cos 45° = -17.828 \text{ m/s}$

$\vec{V}_{RC} = \vec{V}_{RG} - \vec{V}_{CG}$

$V_{RC,x} = -2.475 - 2.828 = -5.303 \text{ m/s}$

$V_{RC,y} = 14.975 - (-17.828) = 32.803 \text{ m/s}$



$|\vec{V}_{RC}| = \sqrt{5.303^2 + 32.803^2} = 33.2 \text{ m/s}$

$\theta = \tan^{-1}\left(\frac{5.303}{32.803}\right) = 9.18°$

Red Skull relative to Captain America

$\vec{V}_{RC} = 33.2 \text{ m/s } [\text{N } 9.18° \text{ W}]$